

Request for Proposals (RFP)

Open-H-Embodiment v2: Healthcare Robotics Data and Annotation Initiative

1 Introduction

Open-H-Embodiment v2 is the second phase of the Open-H-Embodiment initiative, a collaborative effort led by Johns Hopkins University, the Technical University of Munich, NVIDIA, and partner institutions to build open foundations for healthcare robotics. The first release assembled roughly 780 hours of synchronized video and kinematics spanning about 20 robotic platforms and around 50 institutions, and was validated by serving as the training data for the GR00T-H vision-language-action model and the Cosmos-H-Surgical-Simulator world model, as well as many more community-based derivatives.

v2 is a focused push on high-quality, richly labeled data. We annotate procedure recordings against a shared surgical ontology to train commercially usable vision-language models (VLMs), vision-language-action models (VLAs), world-action models (WAMs), and computer-vision (CV) expert models.

The effort centers on two kinds of contribution, and a team may take part through either or both:

- **New medical robotics datasets.** We also welcome newly contributed and extensions of existing datasets pairing synchronized video, robot kinematics, and language annotations.
- **Labeling existing clinical datasets.** The core of the effort: apply the shared ontology to procedure video that has already been collected, producing rich temporal, tool, tissue, and quality annotations.

The primary scope of Open-H-Embodiment includes robotic surgery, robotic ultrasound, and other medical robotics applications. Datasets centered on endoscopy fall within the scope of the sibling Open-H-Endoluminal initiative and should instead be submitted at github.com/open-h/open-h-endoluminal. Contributions are accepted across multiple tiers of realism, including clinical, in vivo, ex vivo, phantom or tabletop, and simulation.

2 Steering Group and v2 Leads

Role	Name	Affiliation
Industry Lead	Dr. Mahdi Azizian	NVIDIA
Surgical Robotics Lead	Prof. Axel Krieger	Johns Hopkins University
Ultrasound Robotics Lead	Prof. Nassir Navab	Technical University of Munich
Clinical Lead	Dr. Filippo Filicori	Northwell Health

3 Objectives and Outcomes

1. **High-quality annotation.** Build a richly annotated subset using a shared surgical ontology (phases, steps, tasks, actions, and gestures) extended with segmentation masks of the anatomy and instruments, motion quality, and intent.
2. **Grow the corpus.** Continue to align leading institutions to contribute synchronized healthcare-robotics data across robotic surgery, robotic ultrasound, and other healthcare tasks.
3. **Weak, scalable labeling.** Develop and apply a weak agentic labeling toolkit that propagates a seed of expert annotations to much larger datasets, so annotation scales beyond what manual labeling alone can reach.
4. **Commercially usable models.** Use the annotated data to train commercially usable VLMs, VLAs, WAMs, and CV expert models.
5. **Open release.** Publish cleaned data, annotations, and model artifacts under a permissive license.

4 Scope of Work

Participation is organized into two tracks. A team may propose to either or both.

4.1 Track A: Robotics Data

Contribute synchronized recordings from one or more of the following groups.

Robotic Surgery. Example skills: needle passing, suture tying, tissue retraction, camera control, surgical dissection, anastomosis, cauterization, irrigation and suction. Platform examples: dVRK, Raven, Virtual Incision MIRA, CMR Versius, and others.

Robotic Ultrasound. Example skills: automated probe placement, sweep acquisition, fetal biometry. Robotic platforms: KUKA LBR Med, UR-series, Franka FR3, and custom cobots with force-feedback probes. Non-robotic contributions that navigate, track, and record medical tasks such as ultrasound imaging are also welcome.

Other Healthcare Robotics. Examples: mobile X-ray positioning, medication dispensing, hospital logistics, patient vital-sign acquisition, imaging robots, and radiation-therapy robots. Platforms: mobile manipulators, collaborative arms, and AMRs.

4.2 Track B: Annotations

For those unable to collect robotics datasets but still eager to have an impact on advancing medical robotics, individuals and groups can contribute annotations on existing procedure data. This track is open to those that wish to contribute their own data that they've already annotated, as well as those who would like to contribute new annotations using our open web-based annotation platform that aims to layer the following on synchronized surgical video:

- **Temporal segmentation:** phase, step, task, action, and gesture, with clean boundaries at the transition between one segment and the next.
- **Segmentation masks:** per-frame masks of the anatomy (for example the cystic duct and cystic artery in cholecystectomy) and of the instruments.
- **Motion quality:** a demonstration-quality rating (for example a 1 to 5 scale).
- **Intent and outcome:** the intended action and, where known, the surgical outcome, including labels for success, recovery, and failure.
- **Reasoning/Narration:** a natural-language description of the sub-task explaining the reasoning of why a particular action/gesture was being performed, or critical highlights and moments during the procedure.

The initial ontology used by our platform spans four procedures, with a shared phases layer and per-procedure tasks: cholecystectomy, inguinal hernia repair, prostatectomy, and hysterectomy.

Annotations can be contributed via our open, web-based annotation platform that we provide, with the ontology, labeling guidelines, and weak-labeling assistance built in. A labeler needs only a laptop with internet access to contribute meaningfully to Open-H-Embodiment. We also accept high-quality annotations produced with equivalent tools.

4.3 Data Requirements and Contribution Tiers (Track A)

Each data proposal must commit to a minimum contribution and to providing time-aligned streams and required metadata.

Time-aligned streams. Provide, synchronized: camera streams (RGB or RGB-D, wrist and room); robot kinematics (pose, force and torque, and other state); and corresponding medical images or video (for example ultrasound B-mode or surgical video).

Kinematics. Native Cartesian end effector kinematics and optionally joint-space kinematics are required for all submitted datasets. However, on a case-by-case basis video-only datasets with inferred poses that demonstrate a high level of accuracy will be considered as well.

Format. Deliver data in the LeRobot data format. Where a robot is involved, supply calibration data and robot CAD and kinematic-tree descriptions (USD, URDF, DH parameters, or equivalent).

Minimum contribution. Each collection setting carries a minimum in hours of synchronized data (below), and a team may contribute across several settings. Higher settings carry a higher relative weight, reflecting their rarity and potential for real-world impact. Preferred order of collection setting: clinical, in-vivo, ex-vivo, phantom or table-top, simulation.

Data Tier	Definition	Minimum Contribution	Relative Weight
Clinical	Collected during approved human procedures with appropriate IRB or ethics oversight.	4 h	×40

In-Vivo	Acquired on living tissue (for example a porcine model) under laboratory or veterinary supervision.	4 h	×20
Ex-Vivo	Performed on isolated tissue samples (for example ex-vivo gallbladder or bowel).	5 h	×4
Phantom / Table-Top	Conducted on physical phantoms, synthetic tissues, or bench-top fixtures.	8 h	×1
Simulation	Digitally simulated using physics-based or differentiable environments.	10 h	×0.2

We encourage proposals to also provide time-aligned narration or description of the sub-task (audio or text), correlated anonymized patient information where available, camera intrinsics, and labels for demonstration quality (expert, intermediate, novice). Additionally, task failures and DAgger-style recovery demos are as valuable as successful demos, thus including tasks success flags with each submitted episode is strongly encouraged (failure, recovery, success).

4.4 Annotation Requirements and Contribution Tiers (Track B)

Annotations are counted in completed labels rather than frames, so a single label that spans many frames (for example a gesture segment or a tool-presence span) counts once. Each label type carries a relative weight reflecting its effort and value.

A contribution is scored as the weighted sum of its labels across types, and must meet an overall minimum of 6,000 weighted labels:

$$\sum (\text{labels} \times \text{relative weight}) \geq 6,000$$

Each row's Minimum Contribution is the count of that type alone that meets this minimum.

Annotation Type	One Label Is	Minimum Contribution	Relative Weight
Tissue segmentation masks	One anatomy mask, tracked through the segment it covers (for example the cystic duct or cystic artery).	1,200 labels	×5
Phase	One phase label covering a complete temporal segment.	1,500 labels	×4

Action/Gesture	One labeled action or gesture segment.	3,000 labels	×2
Tool segmentation masks	One instrument mask, tracked through the segment it covers.	6,000 labels	×1
Verbal narration (audio)	One second of recorded spoken narration of the procedure.	90,000 labels	×0.07

Verbal narration annotators are required to possess at least resident-level medical training or hold an MD (or equivalent) degree, ensuring sufficient expertise to accurately interpret surgical terminology, procedural context, and intraoperative decision-making.

4.5 Focused Tracks and Alternative Formats

To capture the benefits of focused, high-quality datasets while keeping the initiative flexible, we invite groups to propose focused tracks around specific areas, for example a fully annotated cholecystectomy set that demonstrates the ontology end to end, or multi-stitch suturing.

5 Timeline

Milestone	Date
RFP released	July 2026
Proposal submission deadline	September 2026
Data and annotation window	September 2026 to January 2027
Data sample deadline	December 1, 2026
Data cleanup and standardization	January 2027
Model training and validation	February 2027
Public release of dataset, annotations, and models	March 2027

6 Eligibility and Partnership Models

- Academic institutions, start-ups, and industrial healthcare firms are welcome.
- International participation is encouraged.
- Consortia of multiple organizations may submit a joint proposal with a single lead.
- Teams without a robotic platform or their own data may participate through the annotation track (Track B).

Collaboration benefits. All accepted collaborators will be named co-authors on the Open-H-Embodiment v2 dataset publication and on the follow-up paper describing the resulting models. Teams will receive early-access evaluation checkpoints of the models and the dataset itself for their own downstream experiments, and recognition during the v2 dataset and model releases.

7 Proposal Requirements

Proposals for Track B are only required if you are contributing existing labeled datasets. For those wishing to use the hosted annotation site, no submission is needed beyond signing up at annotate.open-h.org.

Submissions (single PDF) must include:

1. Executive summary.
2. Track and target: whether you propose to Track A (data), Track B (annotation), or both; for Track A, the target task groups, robots, sensors, collection environment, and committed hours per collection setting; for Track B, the target procedures, annotation types, collection environment, and committed labels per type.
3. Technical approach: for data, the collection methodology, synchronization, quality assurance, and privacy safeguards (HIPAA and GDPR); for any included labels, the annotation methodology, tooling, ontology alignment, and quality assurance.
4. Team qualifications: key personnel biographies and prior related work.
5. Project plan: work-breakdown structure and Gantt chart aligned with the Section 5 timeline.
6. Data rights and IP statement: confirmation of intent to release data and annotations under the OpenMDW-1.1 license (or compatible) and to comply with steering-group IP policy.

Submit proposals to the [Open-H-Embodiment v2 Google Form](#).

8 Data Governance, Ethics, and Compliance

- **Patient privacy.** All human-subject data must be de-identified to HIPAA Safe Harbor standards or equivalent (GDPR for European contributors), including scrubbing imaging and video metadata on export.
- **Regulatory.** Proposers are responsible for obtaining IRB or ethics approval where required.
- **Annotation provenance.** Annotations should record who or what produced them (expert, trainee, or model-assisted).
- **Licensing.** Final datasets and annotations will be released under the OpenMDW-1.1 license. Synthetic assets must be free of third-party IP encumbrances.
- **Release.** Contributors accept that their datasets will be uploaded publicly to HuggingFace.

9 Contact Information

Administrative questions:

- Sean Huver, shuver@nvidia.com (NVIDIA)
- Axel Krieger, axel@jhu.edu (Johns Hopkins University)

Data, annotation, and engineering questions:

- Vishwesh Nath, vnath@nvidia.com (NVIDIA)
- Juo-Tung (Justin) Chen, jchen396@jhu.edu (Johns Hopkins University)
- Xinhao Chen, xchen254@jhu.edu (Johns Hopkins University)
- Nigel Nelson, nigelN@nvidia.com (NVIDIA)